

Project ID:

R26-IT-123

1. Topic (12 words max)

ShareMate: A Socially Connected Smart Mobility and Community Support App

2. Research group the project belongs to

SST - Software Systems & Technologies

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (300 words max) – references not included in word count.

Urban communities increasingly rely on digital platforms to support daily mobility, shopping assistance, and academic knowledge sharing. However, despite advancements in smart city technologies, many existing applications that provide transportation services, public transport tracking, delivery coordination, and academic content sharing remain fragmented, centralized, and transaction-driven, with limited emphasis on social trust, community collaboration, and affordability [1], [2]. This fragmentation prevents users from benefiting from an integrated digital ecosystem that supports multiple urban needs within the same community.

Public transport live-tracking systems are typically infrastructure-dependent and centrally managed, often resulting in inaccurate or delayed real-time information and a lack of personalization based on user proximity or social connections [6]. Research in urban and geo-social computing highlights that peer-based mobility data and socially connected users can significantly improve real-time accuracy and user trust [1], [3]. Similarly, private vehicle ride-sharing platforms prioritize operational efficiency but fail to support friend-zone-based route sharing, trust modeling, and community-level safety mechanisms, which are critical for sustainable shared mobility [4], [7].

In the domain of shopping and delivery coordination, users continue to depend on commercial services, leading to increased costs, redundant trips, and environmental impact. Studies on the sharing economy emphasize the need for community-driven, trust-based logistics systems to enhance sustainability and reduce unnecessary travel [7], [8].

Additionally, academic note-sharing platforms either allow unrestricted sharing – leading to misuse and lack of sustainability – or rely on expensive subscription models that restrict accessibility for students [10]. This highlights the absence of controlled, fair, and community-oriented academic resource sharing.

The lack of integration across these domains results in higher costs, time inefficiencies, reduced trust, and increased environmental impact. Addressing these gaps requires a geo-social, trust-aware, and community-driven digital platform, contributing to Smart Mobility, Geo-Social Computing, and Community Informatics [1], [9]. Such an approach aligns with SDG 11, SDG 9, and SDG 12 by promoting collaborative resource usage, trust-based interactions, and sustainable urban living [2].

References

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- [2] Batty, M., et al. (2012). Smart cities of the future. *The European Physical Journal Special Topics*, 214(1), 481–518.
- [3] Cho, E., Myers, S. A., & Leskovec, J. (2011). Friendship and mobility: User movement in location-based social networks. *Proceedings of the 17th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 1082–1090.
- [4] Kamargianni, M., Li, W., Matyas, M., & Schäfer, A. (2016). A critical review of new mobility services for urban transport. *Transportation Research Procedia*, 14, 3294–3303.
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- [6] Sampaio, J., & Rossetti, R. J. F. (2015). An integrated platform for real-time public transport information. *Transportation Research Procedia*, 10, 312–321.
- [7] Ganapati, S., & Reddick, C. G. (2018). Prospects and challenges of sharing economy for urban sustainability. *Telecommunications Policy*, 42(1), 77–87.
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- [9] O'Neill, E., Kostakos, V., Kindberg, T., Schiek, A. F., Penn, A., Stanton Fraser, D., & Jones, T. (2006). Instrumenting the city: Developing methods for observing and understanding the digital cityscape. *Proceedings of UbiComp*, 315–332.

- [10] Hilton, J., Wiley, D., Stein, J., & Johnson, A. (2010). The four ‘R’s of openness and ALMS analysis: Frameworks for open educational resources. *Open Learning: The Journal of Open, Distance and e Learning*, 25(1), 37–44.

6. Brief description of Existing Research and Systems (300 words max)

Existing digital platforms addressing transportation, delivery, and information sharing primarily focus on isolated functionalities rather than integrated, community-driven solutions. Ride-hailing services such as Uber, PickMe, and Bolt provide on-demand transportation but operate on a fully commercial and centralized model, lacking social trust mechanisms, route-sharing flexibility, and community collaboration. Although some carpooling systems like BlaBlaCar support shared travel, they do not integrate real-time social grouping, bidding-based seat allocation, or friendzone-based interactions.

Public transport tracking systems, including Google Transit, Moovit, and local transit apps, rely on centralized data sources and fixed schedules. These systems often struggle with real-time accuracy, especially in developing regions where infrastructure support is limited. Furthermore, they do not allow peer-based sharing of live transit data among trusted users.

In the domain of shopping and delivery, platforms such as Uber Eats, PickMe Food, and Instacart focus on paid commercial logistics. Research on collaborative consumption highlights the benefits of peer-to-peer assistance, but most systems lack digital trust verification mechanisms such as receipt-based proof and social trust scoring.

Academic note-sharing platforms like Scribd, Course Hero, and Studocu offer either unrestricted free access or fully subscription-based models. Research shows that unrestricted sharing often leads to misuse, while paid-only models restrict accessibility for economically disadvantaged users.

The key gap across these systems is the absence of a unified, socially connected, trust-based platform that integrates smart mobility, real-time peer transit tracking, collaborative shopping, and controlled educational sharing. Existing solutions treat these domains separately and rarely incorporate geo-social intelligence, friendbased trust modeling, and sustainable access control—limitations that this research aims to address.

7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

ShareMate is a socially connected, trust-based mobile application designed to address key limitations in existing urban digital systems related to transportation, public transit awareness, shopping assistance, and academic resource sharing. Most current solutions operate as isolated services, are largely commercially driven, and lack mechanisms for social trust, peer-powered data sharing, and sustainability. ShareMate introduces a unified geo-social platform that integrates shared public transport tracking, shared private transport coordination, collaborative shopping assistance, and controlled academic note sharing within a single community-driven ecosystem.

The foundation of ShareMate is its friend zone interaction model, which allows users to engage within trusted private circles or broader community networks. Rather than relying entirely on centralized infrastructure, the system enables users to generate and share real-time information socially. This approach improves reliability, reduces operational costs, and enhances user confidence by leveraging familiar social connections. Location awareness and geo-social intelligence allow the platform to dynamically adapt to user proximity, availability, and shared intentions.

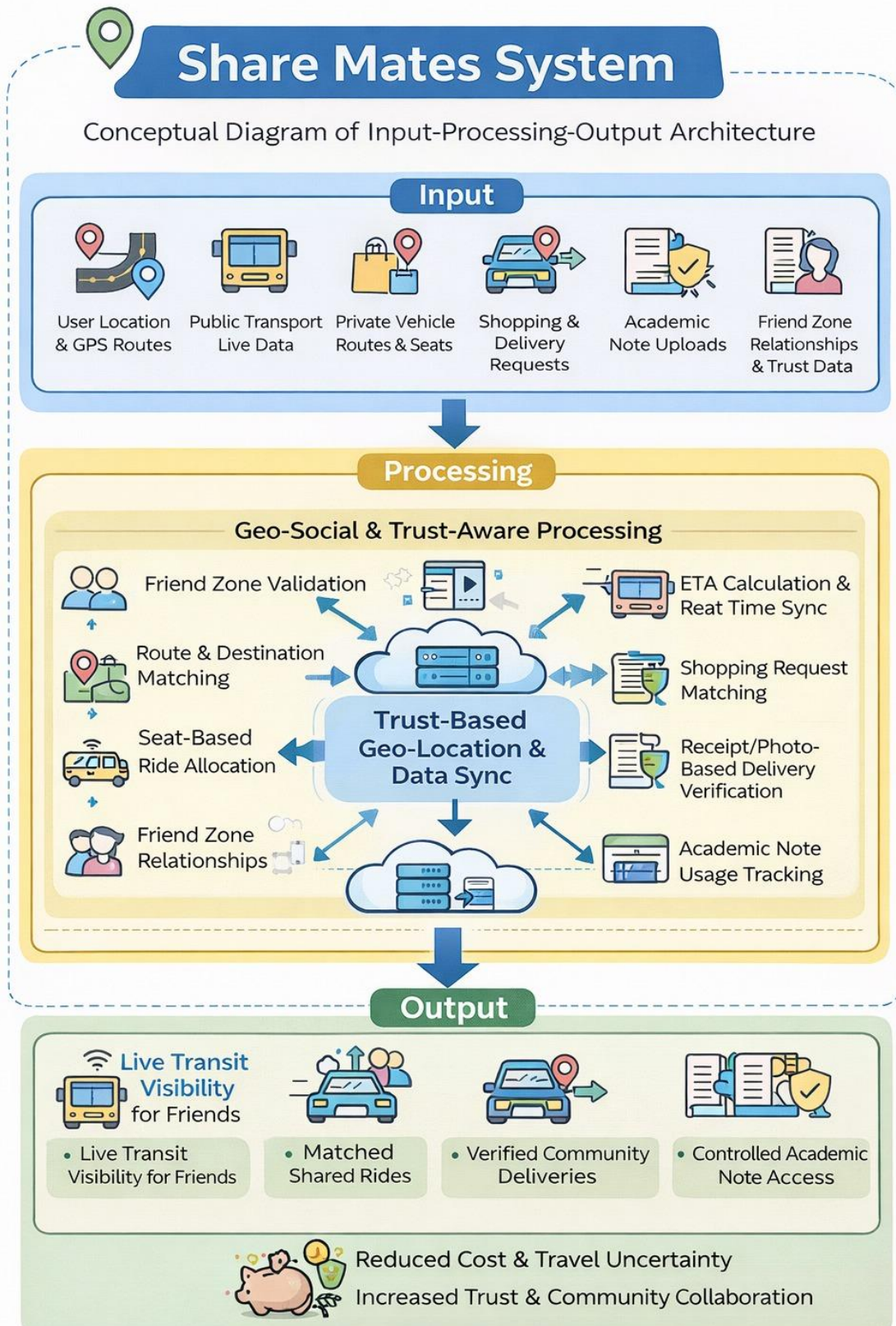
For public transportation, ShareMate enables users traveling on buses or other public vehicles to share live location updates, seat availability, and estimated arrival times with their friend zones. This peer-powered live sharing reduces uncertainty caused by delayed or inaccurate centralized tracking systems and helps commuters make informed travel decisions. In the context of private transport, the application allows users to publish planned routes and destinations, enabling the system to detect overlapping paths and match compatible users within trusted zones. Seat allocation and ride confirmation are managed through social trust and activity history, encouraging safe and cost-effective shared travel.

The shopping and delivery coordination feature supports community-assisted logistics by allowing users to post shopping intentions and delivery requests. Nearby users can fulfill these requests during their own shopping trips, reducing redundant travel and lowering delivery costs. Trust is maintained through receipt or photo-based verification and usage history, promoting accountability and responsible participation. This collaborative approach contributes to more sustainable urban consumption patterns.

Academic note sharing within ShareMate is designed to balance accessibility and sustainability. Students can share learning materials under controlled conditions such as limited access duration, usage tracking, or peer-based permissions. This prevents misuse while encouraging fair and collaborative knowledge exchange, offering an alternative to both unrestricted sharing and costly subscription-based platforms.

From a system perspective, ShareMate follows an input–process–output architecture. User-generated inputs such as location data, routes, transport updates, shopping requests, and academic content are processed through geo-matching, trust evaluation, access control, and real-time synchronization mechanisms. The outputs include matched shared rides, accurate transit updates, verified community deliveries, controlled access to academic resources, trust indicators, and measurable reductions in cost, time, and redundant travel.

Overall, ShareMate represents a community-driven, trust-aware, and sustainable digital solution that enhances urban living by integrating mobility, logistics, and knowledge sharing into a single geo-social platform aligned with smart city principles and sustainable development goals.



8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

The successful design and implementation of the ShareMate mobile application require multidisciplinary expertise covering smart mobility, geo-location technologies, geo-social computing, trust-based digital systems, and mobile application development. The domain knowledge necessary for these areas is widely available through established academic research, open-source technologies, industry documentation, and structured online learning platforms, making the technical and conceptual requirements of the project realistically achievable within an academic setting.

Expertise related to shared public transport systems, friend-based private transport coordination, community-driven logistics, and controlled knowledge-sharing can be developed by engaging with existing research in smart mobility, location-based services, peer-to-peer systems, and community informatics. Reputable academic sources such as IEEE Xplore, ACM Digital Library, and Google Scholar provide extensive literature on public transport tracking, ride sharing, geo-social interaction models, and trust and reputation mechanisms. These resources offer both theoretical foundations and applied methodologies relevant to ShareMate's four core functional domains.

The technical knowledge required for building ShareMate, including mobile application development, real-time data synchronization, GPS and map integration, backend service design, and trust-based access control, is readily accessible. Developer documentation and tutorials for cross-platform frameworks such as Flutter or React Native, along with backend services using Firebase or Node.js, provide clear implementation guidance. Geolocation and routing capabilities can be supported through well-documented mapping services such as Google Maps or Mapbox. These technologies are supported by large developer communities and extensive example implementations, enabling efficient problem-solving and rapid skill acquisition.

The data needed for ShareMate will primarily be generated organically through user interactions with the system. This includes location data for shared transport features, route and destination inputs, public transport live-sharing updates, shopping requests and fulfillment records, academic note uploads, and controlled usage logs. Supplementary non-personal datasets, such as road network data or public transit routes, can be sourced from openly available platforms like OpenStreetMap and government open data portals. During development and testing, synthetic datasets can be created to simulate real-world usage scenarios while avoiding exposure of real user data.

From an ethical and data governance perspective, ShareMate involves the handling of personal and location-based information, necessitating careful attention to privacy and user control. Ethical approval requirements will be followed in accordance with institutional guidelines. The system design incorporates informed user consent, role-based and friend-zone-based data sharing, secure authentication mechanisms, and data anonymization where applicable.

Users retain full control over what information is shared and within which social zones, and the system avoids collecting sensitive personal, biometric, or medical data.

Overall, the required domain expertise, technical skills, and data resources necessary for ShareMate are highly accessible and well-supported. The availability of academic literature, open-source tools, developer platforms, and ethical data-handling practices ensures that the project can be developed effectively, responsibly, and within realistic academic constraints.

9. Objectives and Novelty

Main Objective To design and develop a socially connected, trust-based smart mobility and community support mobile application that enables users to share travel routes, track public transport in real time, coordinate shared shopping activities, and exchange academic resources within friend zones – enhancing affordability, efficiency, safety, and collaboration in daily life.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
IT22054340	Public Transport Live Sharing – FriendZone LiveTransit Share Engine (FLTS Engine)	To design and implement a friend-based real-time public transport sharing system that allows users to share live bus locations, predict arrival times, and pin stops for friends traveling on the same route. • Design UI for live location sharing • Implement GPS-based real-time tracking • Develop friend-zone visibility rules (private/public)	A friend-powered live transit tracking system, where real-time bus locations and ETAs are shared socially instead of relying on centralized infrastructure: A Socially-Connected Real-Time Public Transport Tracking System that allows commuters to share the live location, route progress, and arrival times of buses with their friend zones, enabling friends traveling on the same route to view real-time movement, receive accurate ETAs, and pin bus locations – creating a community-powered, trust-

		• Implement pinning of bus locations and stops • Calculate ETA for each pinned friend location • Integrate push notifications for arrivals • Store and sync live data using a real-time database	based alternative to traditional public transport tracking.
IT22298058	Shared Shopping & Delivery Coordination System – FriendZone SmartShop & TrustVerify Engine (FST)	• Shopping Intention Publishing using Cloud Backend & Structured Data Models Allow users to publish shopping intentions (items, destination, preferred time) using a cloud-based backend (e.g., Firebase/REST API) with structured data storage to enable real-time availability and synchronization among users. • Enable request and acceptance flow, letting friends send shopping requests, accept assistance, and track status (pending, accepted, completed).	Payment-Verified Route & Trust Optimized Shared Shopping: This system introduces a novel approach by combining real-time shopping intention matching, route-aware delivery, and digital payment verification through photo uploads of receipts for both purchased items and delivery completion. It enables community-driven, non-monetary mutual assistance for groceries, food, and daily shopping items, reducing travel redundancy, optimizing delivery efficiency, and fostering trust among users without relying on commercial delivery services.

		Photo-Based Digital Payment Verification: Enables users to upload receipt photos as proof of payment for shopping items and delivery completion, with timestamped records and peer confirmation to ensure transparent and trustworthy verification. Fraud Prevention through Multi-Factor Verification Combine receipt image verification, timestamp validation, and location data to reduce fake submissions and enhance system security.	
IT22088864	Private Vehicle Route Sharing & Friend Zone Ride Engine	- Allow users to share planned travel routes within private and public Friend Zones Using GPS-based route mapping - Enable friends to join shared routes if their destination lies along the same path on the same day - Implement seat-based ride allocation based on vehicle capacity	Friend Zone–Based Route Ownership Transportation Model: This component introduces a novel approach where private vehicle owners initiate shared transportation by publishing their planned routes within trusted Friend Zones. By combining route sharing, seat-aware ride allocation, low-cost pricing, and social bidding, the system eliminates anonymous ride requests and creates a trust-driven, community-based shared mobility model not observed in existing

		• Develop a low-cost destination-based fee mechanism, cheaper than Uber and PickMe • Introduce a flexible price range and bidding system for private and public Friend Zone members Using real-time bidding algorithms, dynamic pricing logic, secure backend transaction handling • Allow drivers(vehicle owner) to select co-travellers based on offered price and available seats • Ensure all ride coordination occurs within trusted Friend Zones	ridehailing platforms or research systems.
IT22215956	A Smart Academic Note-Sharing System with Limited Free Access and Upgrade Mechanisms within the Friend Zone	• Design and implement a user registration and authentication mechanism to uniquely identify users within the system • Develop a note uploading and management module that allows users to share academic notes in digital formats with relevant metadata such as subject and topic	The novelty of this component lies in the usage-based controlled free access model for academic note sharing . Unlike existing platforms that offer either unrestricted free access or fully paid access, the proposed system introduces a limited free-access threshold, allowing users to access notes only five times before requiring an upgrade.

		• Implement a usage-tracking mechanism to monitor the number of notes accessed by each user • Enforce a controlled free-access policy by allowing users to access shared notes only up to five times • Restrict further access automatically once the free usage limit is exceeded • Design and integrate an upgrade mechanism that enables users to continue accessing notes after completing the free usage quota • Update user access levels dynamically based on upgrade status and notify users upon successful access changes	This approach transforms conventional note-sharing platforms into a sustainable and fair access controlled system, balancing learning accessibility with responsible usage and monetization.
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**10. This part is to be filled by the
 Supervisor and the Co-supervisor of the Project.**

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes	☒	No	☐
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b) The proposed topic exhibits novelty.

Yes	☒	No	☐
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c) The student group can successfully execute the proposed project.

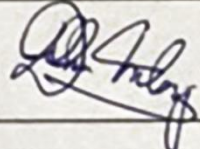
Yes	☒	No	☐
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d) The proposed sub-objectives reflect the students' areas of specialization.

Yes	☒	No	☐
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e) Any other comments:

N/A

	Title	First Name	Last Name	Signature
Supervisor	Prof.	Dilshan	De Silva	
Co-Supervisor				
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				